Ollscoil na Gaillimhe
University of Galway

Distance, Mobility, and Mode Choice: A Spatial Analysis of School Travel Behaviour in Ireland

Judith Beltrán López, Sananthaa Jagadesan Senthamaraikannan[1]

Background and Problem

Daily travel to school plays an important role in shaping physical activity levels among children and adolescents. In recent years, increased reliance on unhealthy modes of transport, such as private cars, has reduced opportunities for active travel, including walking and cycling. In Ireland, factors such as school location, travel distance, and neighbourhood characteristics strongly influence whether students use healthy or unhealthy ways of travelling to school.

Reduced physical activity linked to unhealthy transport choices contributes to rising levels of overweight and obesity in the population. Encouraging active school travel is therefore a key public health concern, particularly as travel habits formed during childhood often persist into adulthood.

Objectives of Project

1. Map and analyse the spatial distribution of schools across Ireland.
2. Examine the distance between students' residential locations and their schools, and investigate student mobility related to school attendance.
3. Analyse school travel behaviour and assess the prevalence of healthy versus less healthy commuting modes.
4. Explore the relationship between school proximity, travel mode, and potential health implications.

Data Sources and Datasets

Data on schools in Ireland are of the following Types:

1. School-Level Data (Department of Education) School-level information for schools in Ireland, including school type, enrolment size, and location details (address, Eircode, latitude, and longitude). These data allow schools to be mapped and linked to geographic areas. (https://www.gov.ie/en/department-of-education/collections/data-on-individual-schools/)

School enrolment data were treated as point-level observations and kriged to produce a continuous surface highlighting spatial patterns in school size.

2. Settlement-Level Travel to Education Data (Central Statistics Office) Aggregated census data describing how people travel to education (e.g., walking, cycling, public transport, car) at settlement level. Data are available for multiple census years and include all forms of education and childcare, with some tables providing breakdowns by education level.

https://data.cso.ie/table/SAP2022T11T1TOWN22 (2022 data - with all forms of education and childcare combined for ~800 settlements) https://data.cso.ie/table/SAP2016T11T1SET (2016 data - with all forms of education combined for ~800 settlements) https://data.cso.ie/table/E6013 (2016 data - with data separate for different periods of education for ~200 settlements)

Census data were analysed at settlement level using predefined spatial boundaries to examine patterns in travel to education across Ireland.

3. Spatial Boundary Data (CSO) Geographic boundary files for towns and urban areas in Ireland. These define the spatial extent of settlements and are used to map and link travel behaviour data to specific locations. https://www.cso.ie/en/census/census2022/census202

Early Results

As a first step in the analysis, we focused on understanding the data we are working with. To achieve this, we explored the spatial distribution of schools across Ireland. By mapping school locations and producing a density plot, we were able to visualise areas with high and low concentrations of schools.

This initial exploratory analysis helps to validate the dataset, identify spatial patterns, and highlight regional differences in school availability. Understanding where schools are clustered provides an essential foundation for subsequent analyses on school accessibility, travel distances, and the use of healthy versus unhealthy modes of transport to school.

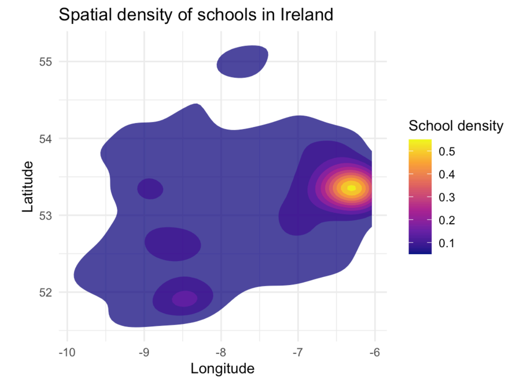

Figure 1: Great figure!

Overall, the graph highlights a strong urban–rural divide in the spatial distribution of schools. Identifying Dublin as the main density hotspot is important, as such concentration is likely associated with shorter travel distances and a greater potential for active modes of transport, while lower-density regions may rely more heavily on motorised transport options due to longer travel distances.

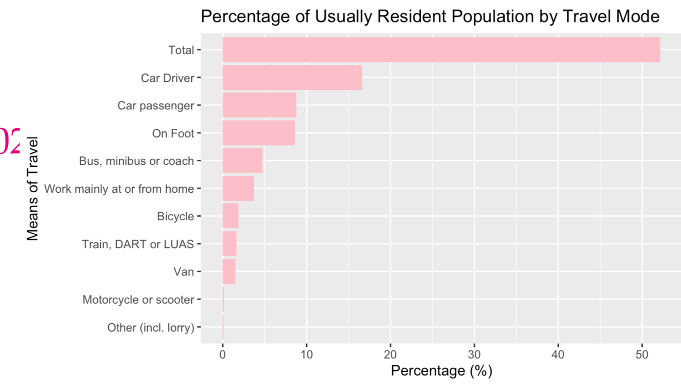

Figure 2: Great figure!

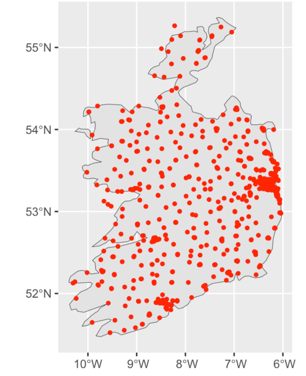

Figure 3: School Population

Kriging is a geostatistical interpolation method that uses spatial dependence to predict values at unsampled locations, making it useful for identifying spatial patterns and inequalities related to school locations, travel behaviour, and potential health impacts.

```
## [using ordinary kriging]
```

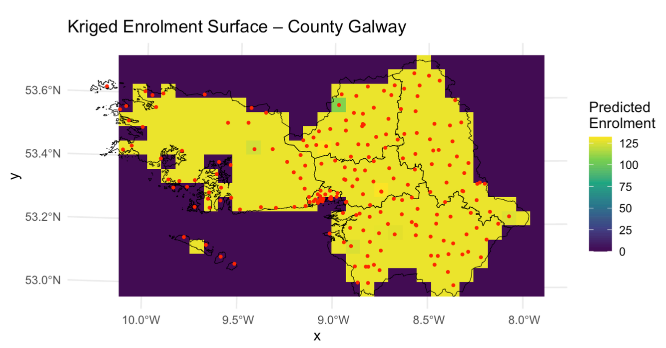

Figure 4: Resident Population by Travel Mode

The kriged surface shows higher predicted enrolment in areas where schools (red points) are more densely clustered, particularly around urban centres, while lower values appear in more rural and sparsely served areas. This highlights spatial inequality in school enrolment across County Galway.

Next Project Steps

Define Outcome Variables: Clearly decide what outcomes will be studied (e.g., exam results, enrollment changes, or resource availability) and ensure they are clearly measured and relevant to the research questions.

Control for Confounding Factors: Identify factors such as socioeconomic background, location, school size, and demographics that may influence results, and account for them in the analysis.

Further Statistical Analysis: Extend the exploratory analysis by applying regression or multivariable models to examine relationships while adjusting for other influencing factors.

GitHub

The code and datasets for this project can be viewed at our GitHub repository here: https://github.com/judithbeltranlopez/Final-Project

References